



YOLOV8 FOR ROAD ACCIDENT DETECTION: A VISION-BASED DEEP LEARNING MODEL

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- 1. Abstract**
- 2. Problem definition and Solution**
- 3. Literature Review**
- 4. Dataset**
- 5. Methodology**
- 6. Experimental Results**
- 7. Future Works & Conclusion**
- 8. References**
- 9. Demo**

Abstract

The paper compares YOLOv8 and Two-Stream CNN architectures for automatic accident detection in Bangladesh. Both models use bilateral filtering to minimize noise and preserve edge information. A personalized dataset was developed, and YOLOv8 performed better with 83.3% test accuracy compared to Two-Stream CNN's 74.0%. YOLOv8 also achieved better validation accuracy and class-wise metrics, with a higher precision and recall for Class 0 and Class 1 scenes. Despite having more parameters, YOLOv8 achieved the highest performance at 59 epochs, reaffirming its importance in accident detection systems.

Problem Definition and Solution Approach

Problem

- 1.35M road fatalities annually worldwide (WHO)
- 105,000+ deaths in Bangladesh (2014-2024)
- Traditional detection: delays, human limitation, inconsistent monitoring

Solution

- Real-time automated accident detection via deep learning
- Comparing YOLOv8 and Two-Stream CNN methods
- Improved speed, accuracy, and coverage



Literature Review



Paper & Publication Year	Author	Methodology	Accuracy	Limitations
A Deep Learning based Accident Detection System(2020)	Gokul Rajesh, Amitha Rossy Benny, Harikrishnan A, James Jacob Abraham, Nithin Prince John	CNN-based accident detection using image processing and frame averaging; custom dataset; GSM alert system.	85%	Dataset limited to Indian accident scenarios; 85% accuracy leaves room for improvement; reliance on fixed cameras in accident-prone areas.
A Novel Dataset for CCTV Traffic Camera based Accident Analysis(2018)	Ankit Parag Shah, Jean-Bapstite Lamare, Tuan Nguyen-Anh, Alexander Hauptmann	Created CADP dataset; Faster R-CNN with Context Mining; Accident LSTM for forecasting.	85%	Focus on fixed traffic cameras only; small pedestrian detection still challenging; legal/data access constraints limit dataset diversity.
Accident Detection System Using YOLOv8 and CNN Algorithms(2023)	Aadarsh Darokar, Akanksha Talware, Amit Jadhav, Darshana Guntiwar, Swati Dandhare, Prof. Devendra Dandekar	YOLOv8 (CSPDarknet53 backbone) with CNN-LSTM hybrid; trained on labeled accident images/videos.	Not explicitly reported; stated improved real-time detection accuracy	Requires high-quality annotated data; performance sensitive to lighting/weather; computationally intensive for real-time use.

Literature Review

Paper & Publication Year	Author	Methodology	Accuracy	Limitations
Accident Detection Using Convolutional Neural Networks(2019)	Sreyan Ghosh, Sherwin Joseph Sunny, Rohan Roney	Inception v3 CNN + LSTM for video frame classification; deployed on Raspberry Pi with GSM alerts.	95%	Limited to highways with sparse traffic; needs large labeled datasets; hardware limits due to Raspberry Pi constraints.
Accident Detection using Deep Learning: A Brief Survey(2020)	Renu, Durgesh Kumar Yadav, Iftisham Anjum, Ankita	Literature survey of CNN, LSTM, sensors, GSM, GPS based accident detection methods.	Survey paper — no accuracy	Reviews gaps but no new solution; limited by available research; rapid tech changes may date some info.
Comparison of YOLO Algorithms for Vehicle Accident Detection and Classification(2024)	Ahmed N. Nusari, Abdulfattah E. Ba Alawi, Ferhat Bozkurt, İbrahim Yücel Özbek	Comparative evaluation of YOLOv6, YOLOv7, YOLOv8, YOLOv9 on accident datasets.	YOLOv8m precision 97.9%; YOLOv9e mAP50-95: 94.1%	Limited dataset diversity; dataset quality variability affects results; real-world deployment challenges not fully addressed

Literature Review



Paper & Publication Year	Author	Methodology	Accuracy	Limitations
Driver Decision Support System based on Driver Behaviour Prediction to Avoid Accidents in Intelligent Transport System(2024)	Balasubramani S, John Aravindhar D, P.N. Renjith, K. Ramesh	Uses K-Means clustering to classify driver behavior (normal/abnormal) based on physical behavior detection (face monitoring, alcohol sensor) and external parameters (speed, lane change, weather, traffic). Machine learning models (KNN, SVM, Decision Tree, Logistic Regression, Naïve Bayes) evaluated for prediction.	KNN achieved best accuracy: 92% approx.; Precision 99%; Recall 98%; F1-Score ~98%. SVM accuracy 90%.	Dataset generated using SUMO simulation; real-world deployment and sensor errors not deeply explored; model performance dependent on sensor accuracy and traffic scenario complexity.
Edge Detection Technique Based on Bilateral Filtering and Iterative Threshold Selection Algorithm and Transfer Learning for Traffic Sign Recognition(2023)	Milind Parse, Dhanya Pramod	Proposed edge detection algorithm using bilateral filtering with adaptive thresholding and varying aperture size. Combined with transfer learning on CNN models (VGG-16, Inception V3) for traffic sign recognition. Evaluated on GTSRB dataset.	Testing accuracy improved from 91.78% to 94.32% (VGG-16) and from 98.67% to 99.49% (Inception V3) after edge detection pre-processing.	Limited to traffic sign recognition; tested mainly on GTSRB and custom dataset; real-time performance and hardware implementation details not addressed.

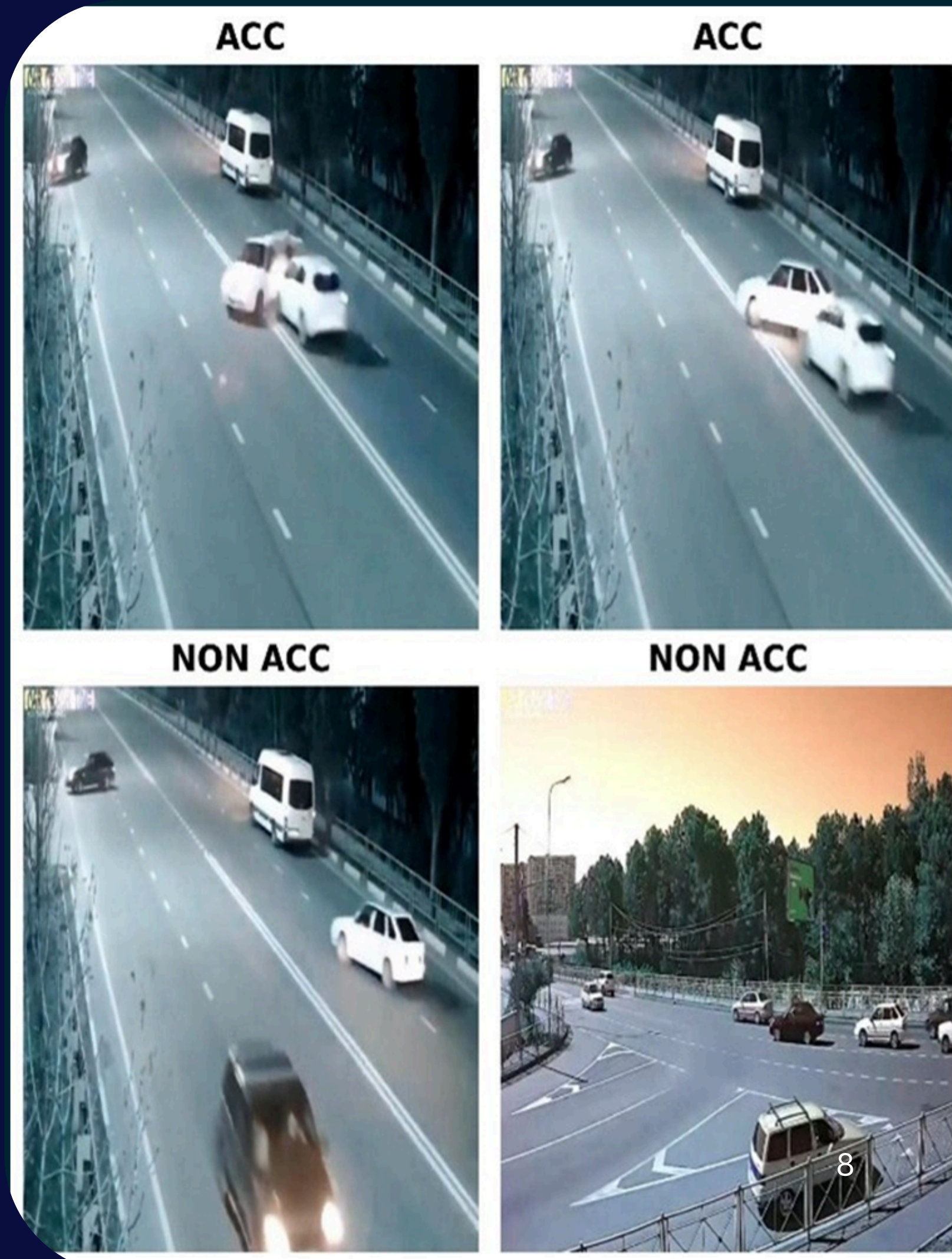
Dataset

Source :

Multiple sources including GitHub, public datasets, and CCTV footage

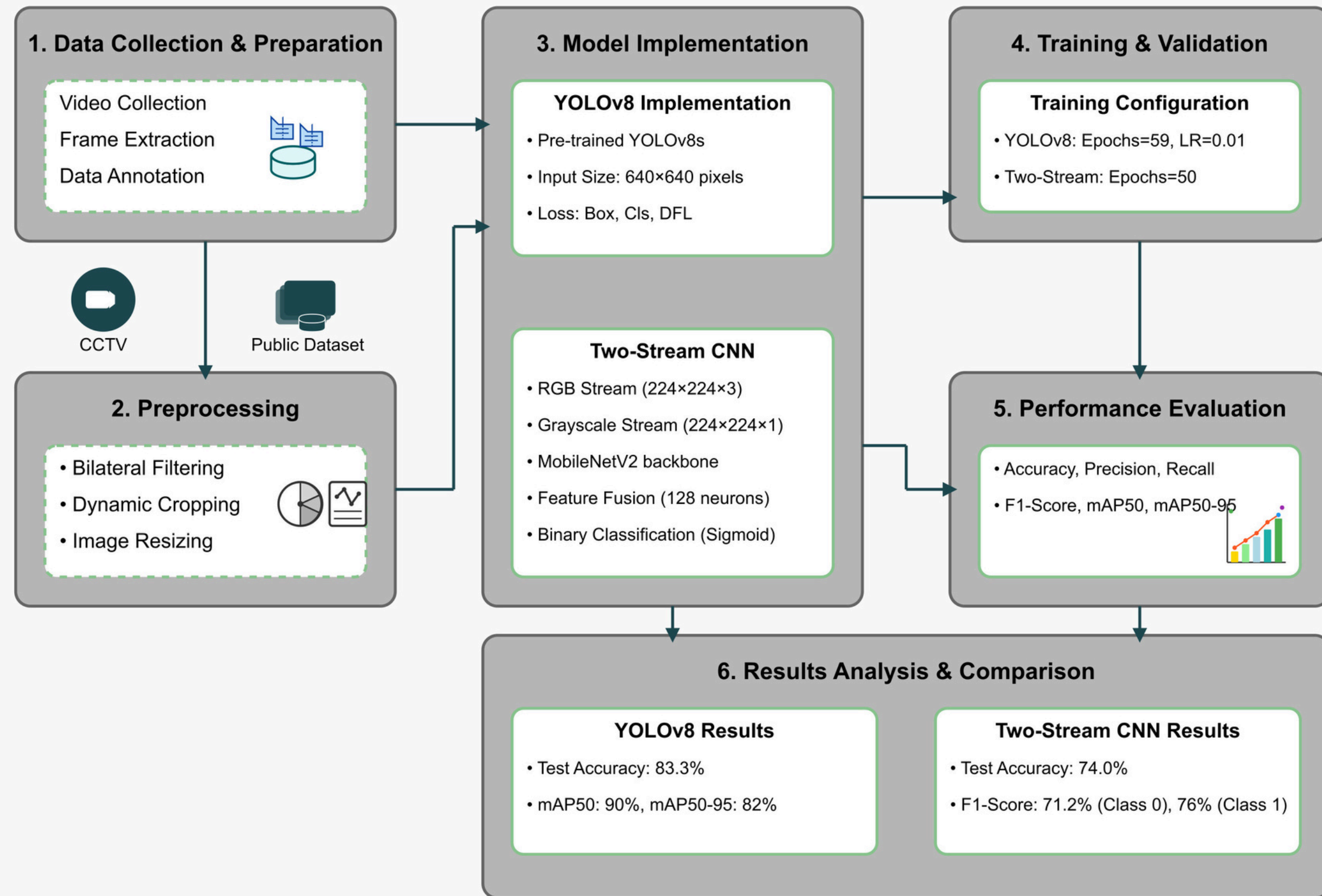
Size: Approximately 4000 images total

Classes: Binary classification (accident and non-accident)



Methodology

METHODOLOGY



IMPLEMENTATION TOOLS

Category	Tool/Technology
Programming Language	Python
Deep Learning Framework	PyTorch (YOLOv8), TensorFlow/Keras
Web Framework	Flask
Frontend Technologies	HTML, CSS, JavaScript
Image Processing	OpenCV
Data Annotation	Roboflow
GPU	NVIDIA Tesla T4 GPU

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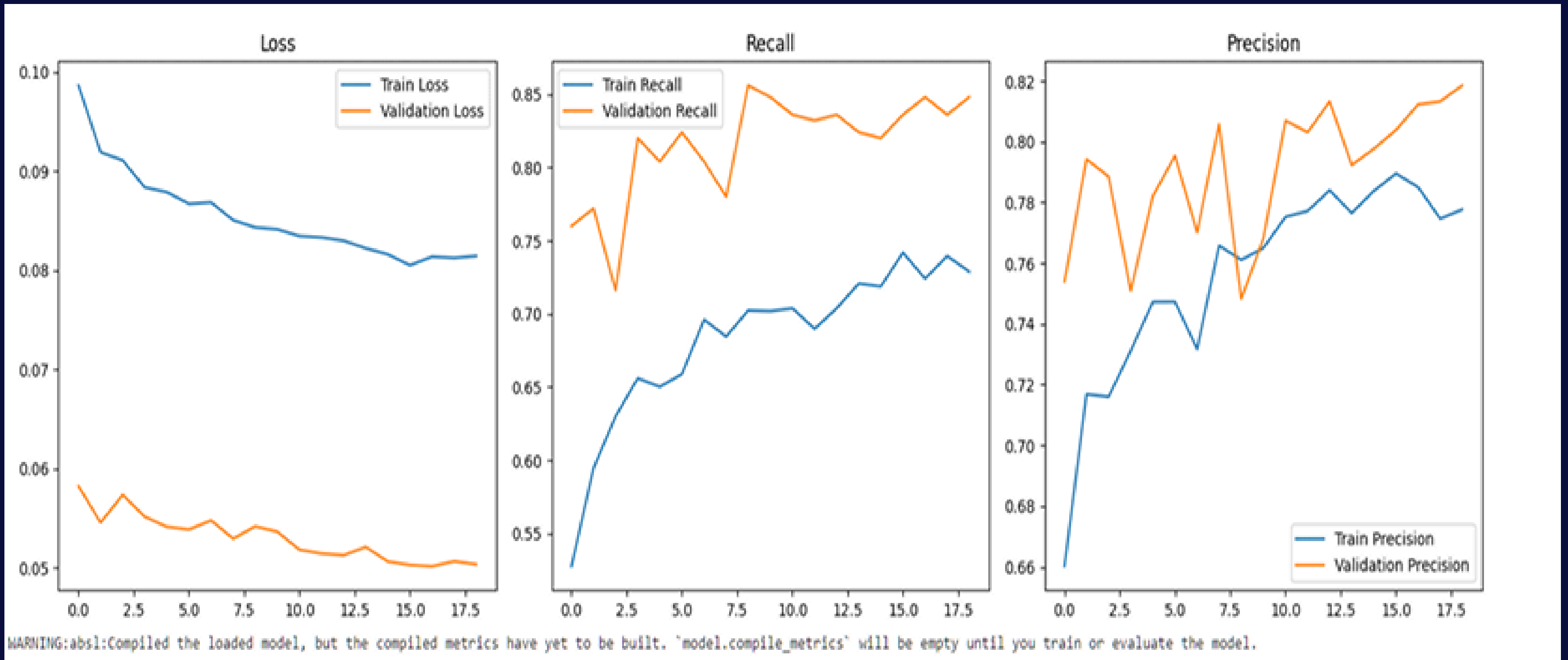
def __init__(self):
    self.file = None
    self.fingerprints = set()
    self.logdups = True
    self.debug = debug
    self.logger = logging.getLogger(__name__)
    if path:
        self.file = open(os.path.join(path, 'requests.log'), 'a')
        self.file.seek(0)
        self.fingerprints.update(e.request() for e in self)

    @classmethod
    def from_settings(cls, settings):
        debug = settings.getbool('SUPERFILTER_DEBUG')
        return cls(job_dir(settings), debug)

    def request_seen(self, request):
        fp = self.request_fingerprint(request)
        if fp in self.fingerprints:
            return True
        self.fingerprints.add(fp)
        if self.file:
            self.file.write(fp + os.linesep)

    def request_fingerprint(self, request):
        return request_fingerprint(request)
```


Training and Validation Performance Analysis



Results Comparasion

Metric	YOLOv8	Two-Stream CNN
Test Accuracy	0.833	0.740
Validation Accuracy	0.912	0.806
Precision	Class 0 : 0.79, Class 1 : 0.90	Class 0 : 0.872, Class 1 : 0.659
Recall	Class 0 : 0.77, Class 1 : 0.94	Class 0: 0.60, Class 1: 0.897
F1-Score	Class 0 : 0.78, Class 1 : 0.92	Class 0: 0.712, Class 1: 0.76
mAP ₅₀	0.90	N/A
mAP ₅₀₋₉₅	0.82	N/A
Parameters	11,126,358	2,422,083
Optimal Epoch	59	30 (Including Fine-Tuning)

Limitations



Dataset Bias

Limited accident scenarios and traffic diversity may overfit models.



Environmental Factors

Reduced accuracy under fog, rain, and poor nighttime lighting.





Future Work

Hybrid Models

Combine YOLOv8's speed with Two-Stream CNN's temporal accuracy for improved detection.

Robustness Improvement

Develop adaptive models to handle environmental changes, regional variations.

Driver Behavior Analysis

Integrate driver behavior prediction for a more proactive accident detection system.

References

1.

G. Rajesh, A. R. Benny, H. A., J. J. Abraham, and N. P. John, "A Deep Learning based Accident Detection System," International Journal of Engineering Research & Technology (IJERT), vol. 9, no. 4, pp. 123-130, 2020.

2.

A. P. Shah, J.-B. Lamare, T. Nguyen-Anh, and A. Hauptmann, "A Novel Dataset for CCTV Traffic Camera based Accident Analysis," IEEE International Conference on Computer Vision Workshops, pp. 3500-3510, 2018.

3.

A. Darokar, A. Talware, A. Jadhav, D. Guntiwar, S. Dandhare, and D. Dandekar, "Accident Detection System Using YOLOv8 and CNN Algorithms," International Journal of Computer Applications, vol. 182, no. 45, pp. 18-25, 2023.

4.

S. Ghosh, S. J. Sunny, and R. Roney, "Accident Detection Using Convolutional Neural Networks," IEEE International Conference on Machine Learning and Applications, pp. 110-115, 2019.

5.

Renu, D. K. Yadav, I. Anjum, and A. (Last name unknown), "Accident Detection using Deep Learning: A Brief Survey," Journal of Emerging Technologies and Innovative Research, vol. 7, no. 3, pp. 58-66, 2020.

6.

A. N. Nusari, A. E. Ba Alawi, F. Bozkurt, and İ. Y. Özbek, "Comparison of YOLO Algorithms for Vehicle Accident Detection and Classification," Sensors, vol. 24, no. 1, pp. 1-15, 2024.

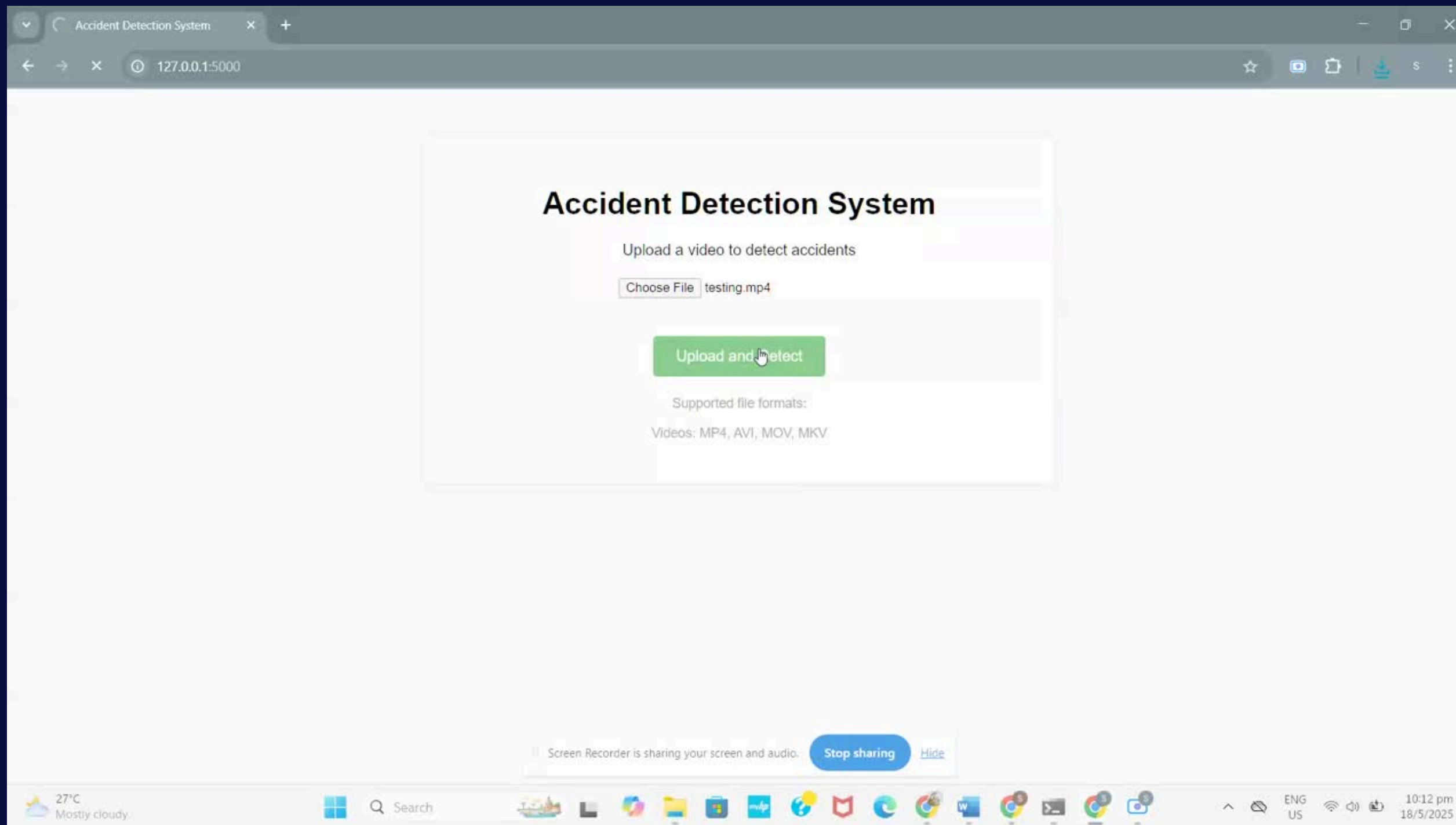
7.

B. S., J. Aravindhar, P. N. Renjith, and K. Ramesh, "Driver Decision Support System based on the Driver Behaviour Prediction to Avoid Accidents in Intelligent Transport System," International Journal of Cognitive Computing in Engineering, vol. 5, pp. 1-13, Dec. 2023, doi: 10.1016/j.ijcce.2023.12.001.

8.

M. Parse and D. Pramod, "Edge Detection Technique Based on Bilateral Filtering and Iterative Threshold Selection Algorithm and Transfer Learning for Traffic Sign Recognition," Scientific Journal of Silesian University of Technology. Series Transport, vol. 119, pp. 199-222, 2023.

Demo



Thank
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